

Tristan David Carter

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Education

Duke University

Aug 2025 – May 2029

B.S.E. Electrical & Computer Engineering (ML Concentration), B.S. Computer Science, GPA: 3.837/4.0

Relevant Coursework: Linear Algebra, Data Structures & Algorithms, Computer Architecture, Signals & Systems

Projects and Technical Experience

Multimodal Time-Series Forecasting of Alzheimer's Disease Progression

Jan 2026 – Jul 2026

- Built the longitudinal data pipeline for a multimodal Alzheimer's forecasting system, migrating MRI/PET cohorts to AWS S3 and automating BIDS conversion, skull stripping, longitudinal and cross-modality registration, quality control, and harmonization to prepare irregular patient histories for downstream modeling.
- Using the resulting longitudinal representations, co-developed continuous-time progression models with optical flow, Optimal Transport flow matching, and a JAX/Diffrax Neural CDE to model anatomical and cognitive trajectories; incorporating clinical features reduced validation ADAS-13 MAE by approximately 50% compared with trajectory-only models.

Duke WATCH: AI Scraper Detection & LLM Attribution — Duke Code+ Intern

May 2026 – Aug 2026

- Built and deployed an AWS honeypot platform (FastAPI/PostgreSQL/Docker), using Ansible to provision multiple sites at scale behind SNI-routed Nginx and serve each visitor content generated by a LoRA-fine-tuned Qwen3-1.7B with embedded per-device canary tokens; queried target LLMs and scored token matches with RapidFuzz to detect likely ingestion of scraped content, with Prometheus/Grafana/Loki for monitoring.
- Built the complementary scraper-attribution pipeline, combining browser and behavioral fingerprints with passive Zeek JA4/JA4T network signals and cross-layer consistency checks to link rotating IPs, user agents, and ASNs to recurring devices; clustered coordinated scraper campaigns using Louvain community detection and GraphSAGE.

ML Damping and Forcing Estimation in Spring–Mass Systems, 2024 AIAA YPSE Conference

June – Nov 2024

- Built DNN/LSTM models to estimate damping and forcing parameters from displacement data with $\leq 5\%$ gaussian noise; validated against Runge–Kutta/Laplace solutions and McPherson-strut simulations.

Optimizing Adaptive Cruise Control in Adverse Environments

Sept 2023– Aug 2024

- Modeled Mazda CX-90 adaptive cruise control with nonlinear dynamics and Lyapunov/sliding-mode control, applying Chebyshev Type II filtering to denoise radar returns; paired the controller with a YOLOv5 detector trained on the Canadian Adverse Driving Conditions dataset to improve perception robustness under snow and rain.

Full-Stack Database Management Application, National FBLA Conference 2024

Dec 2023 – July 2024

- Designed and implemented a full-stack web app with Java (JSP, Servlets, JDBC) and MySQL, deployed on AWS EC2 with Apache Tomcat as a scalable cloud-hosted solution.

Journal Papers

Quasi Monte Carlo Simulations, Howard County Math Journal ↗

Aug 2025

Sobol sequences, Xorshift generation, numerical integration, and Asian option pricing under Geometric Brownian Motion.

Parameter Estimation for a Linear System of Equations, Howard County Math Journal ↗

Aug 2024

Matrix inversion, least-squares, and minimum-norm parameter estimation.

Technical Stack

Languages: Python, C++ , Java, JavaScript, SQL, Unix/Bash, HTML/CSS

ML/AI Stack: PyTorch, TensorFlow, JAX, scikit-learn, NumPy/SciPy/Pandas, Hugging Face, PEFT/LoRA, MuJoCo

Backend/Database: FastAPI, Flask, Uvicorn, REST APIs, Nginx, PostgreSQL, SQLite

Infrastructure: AWS (EC2, S3, IAM), Terraform, Ansible, Linux, Git, CI/CD, Prometheus, Grafana, Loki, PromQL

Networking/Security: TCP/IP, TLS/TCP fingerprinting, Zeek, ACME, bcrypt

Extracurricular

Duke Swim Instructor (25-26), Duke Electrical Vehicles (25-Present), Duke Quantum Information Society Executive Board (25-Present), Duke Applied ML Club (26-Present), Cornell University RL Seminar (26-Present)